

Figure 4: *coap-request* node in NodeRED

requests to the CoAP server. This node takes input or gives output in the form of a payload property of a JavaScript object. Currently, this node supports CON messages.

The npm package *coap-cli* provides a simple command line tool for making various requests with the following syntax:

```
coap <request> <options> URI
```

A few options are: *-p* for payload, *-o* for observe, *-c* for CON and *-n* for NON.

e.g.: *coap post coap://localhost/updates! -p hello #for POST operation*

coap -o coap://localhost/RtcData #for observe operations

Other implementations are:

- libcoap and freecoap in C, suitable for porting on the lwip stack
- nCoAP in Java
- npm packages *coap*, *node-coap*, *coap-cli*, etc, in Node.js
- CoAPthon, *aiocoap*, *txthings*, etc, in Python
- ESP8266 libraries

Stay tuned to <http://coap.technology/impls.html> for frequent updates.

Packet sniffing using Wireshark

Wireshark comes with support for dissecting CoAP messages, and the final version of the protocol is well supported from version 1.12 onwards. For this, install Wireshark from the package manager or build it from source. Then run Wireshark with suitable privileges and start capturing by selecting the interface on which the message flow exists. Type *coap* in the filter box and hit *Enter* to eliminate other packets. In each message, you can check various header elements, tokens, options and payloads, if any. For example, request the URL *coap://localhost/Temperature* using the *GET* method, and check

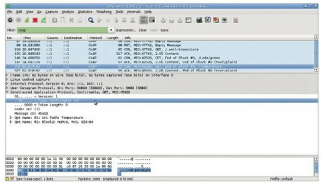


Figure 5: Protocol debugging with Wireshark

if it is of the CON or NON type, message ID, Uri-Path option, etc. You can also find a piggy backed response with the same message ID of request, content format as *text/plain*, response code 2.05 and suitable payload. Now, select the NON type from behaviour drop-down in the Copper plugin and try the same request. You will find an independent NON message with a different ID as the response. You can dissect the message flow in Ping, discover operations and other request methods.

You can analyse capture files in the *pcapng* format recorded from Wireshark or an equivalent, using the *coap.me* pcap interpreter. It also provides a crawler client, which crawls though any public CoAP server using *coap.me*.

Android apps

Any IoT application is not complete without a mobile app, and Android has good support for this. A prebuilt application *Aneska* from Playstore is available. You can develop custom apps by building code available from Eclipse Californium demo apps (*cf-android*) or SpitiFirefox from github.com/okleine using Android Studio or the Gradle build system.

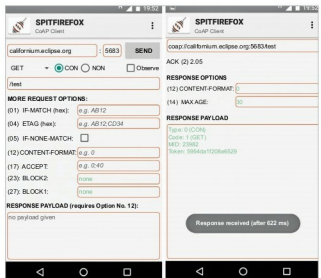


Figure 6: SpitiFirefox Android app